**Unit:** 6 Unit 6: Systems of Linear Equations and Inequalities  **Name:** ____________
**Topic:** Solving Systems by Graphing  **Date:** __________

## Advanced (Habanero)

**Q1.** What is the $x$-coordinate of the point of intersection between the lines $y = 2x + 1$ and $y = x - 3$?
(A) $-2$
(B) 1
(C) 2
(D) 4
(E) 6

**Q2.** If the system of equations $x + y = 4$ and $2x - 2y = -4$ has a solution, what is the $y$-coordinate of the solution?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4

**Q3.** Which of the following systems of linear equations has no solution?
(A) $x + y = 2$ and $x + y = 3$
(B) $x - y = 1$ and $x + y = 1$
(C) $2x + 2y = 4$ and $x + y = 2$
(D) $3x - 2y = 1$ and $x + 2y = 3$
(E) $x - 2y = -3$ and $2x + 4y = 6$

**Q4.** The system of equations $y = 2x - 1$ and $y = x + 1$ has a solution at $x =$
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4

**Q5.** The point of intersection between $y = x^2$ and $y = 2x$ is
(A) $(0, 0)$
(B) $(1, 1)$
(C) $(1, 2)$
(D) $(2, 4)$
(E) $(4, 8)$

**Q6.** The system of linear equations $x + 2y = 4$ and $2x + 4y = 8$ has
(A) a unique solution
(B) no solution
(C) infinitely many solutions
(D) two distinct solutions
(E) three distinct solutions

**Q7.** If the system of equations $y = 2x + 1$ and $y = x - 2$ has a solution, what is the $x$-coordinate of the solution?
(A) $-1$
(B) 1
(C) 2
(D) 3
(E) $-3$

**Q8.** The point $(1, 3)$ satisfies which system of equations?
(A) $x + y = 4$ and $2x - 2y = -2$
(B) $x - y = -2$ and $2x + y = 5$
(C) $x + y = 2$ and $2x - 2y = 0$
(D) $2x + 3y = 9$ and $x - y = 2$
(E) $3x + 2y = 7$ and $2x - 3y = -5$

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**Open Ended Questions (FRQ)**

**Q9.** Solve the system of linear equations $x + 3y = 7$ and $2x - 2y = 2$. Show all work and provide the solution as an ordered pair $(x, y)$. Provide your answer in the space below.

**Q10.** Graph the system of linear inequalities $y \leq x + 1$ and $y \geq 2x - 3$. Identify the vertices of the solution region and provide the equations of the boundary lines. Provide your answer in the space below.

**Chef's Tips**



- Tip 1: Please ensure students have graph paper for the free response questions.
- Tip 2: Calculators may not be used for this assessment.
- Tip 3: Emphasize the importance of labeling and clearly showing all work for the free response questions.